

Silent Suspense

Multimodal Haptics for Horror Film Experiences for Deaf and Hard-of-Hearing (DHH) Audiences

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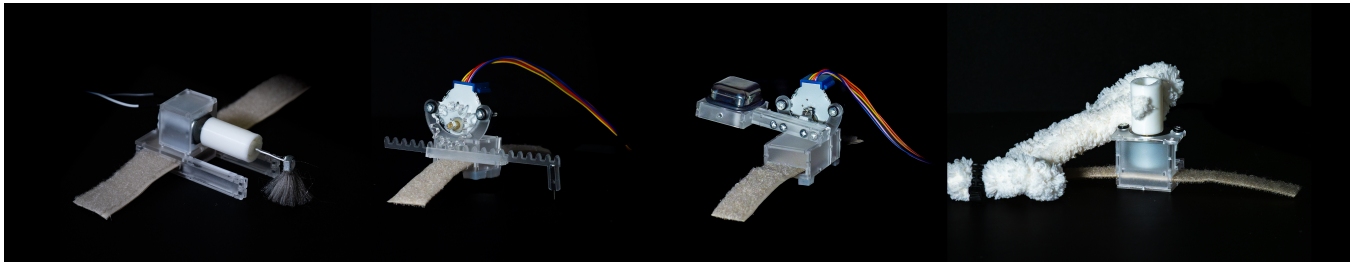


Figure 1: Four haptic devices that convey suspense in horror films through texture, locomotion, temperature, and pressure.

Abstract

We introduce Silent Suspense, four wrist-worn haptic devices designed to translate horror film suspense into tactile sensations for Deaf and Hard-of-Hearing (DHH) audiences. Unlike prior vibration-based approaches for auditory accessibility, our devices aim to deliver the emotional tension of horror through embodied sensations in different tactile modalities—*texture*, *locomotion*, *temperature*, and *pressure*. A user study with 12 DHH participants showed that all devices improved narrative engagement of the horror movie, each device triggering specific emotional responses. Our work contributes by opening a design space for cinematic haptics, demonstrating how diverse tactile modalities can effectively convey suspense to widen horror-film accessibility.

CCS Concepts

• **Human-centered computing** → **Accessibility systems and tools**.

*Three authors contributed equally to this research.

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Keywords

Accessibility, Haptics, Horror film experience, Multimodality

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1 Introduction

Horror films derive much of their scariness from suspense, a feeling of anxious anticipation that builds as audiences wait for something frightening to happen [13]. A huge part of creating this suspense comes from the audio—the background music and sound effects [12, 14]. These sounds create the film's rhythm of suspense, guiding audiences through the narrative's pace and building anticipation for emotional climaxes [2, 8]. However, for Deaf and Hard-of-Hearing (DHH) audiences, this crucial layer of sound-driven immersion remains largely inaccessible [18]. They often struggle to fully engage with horror films because they cannot experience the emotional tension that sound provides. While current accessibility efforts include audio descriptions in subtitles [1], simply reading what's happening on screen and what kind of audio is playing cannot effectively convey the tense atmosphere that audio creates in horror films [5, 16]. To address this issue, we propose leveraging various tactile feedbacks, which have proven effective in augmenting music

	Texture	Locomotion	Temperature	Pressure
Narrative understanding	4.17	3.17	3.17	4.17
Attentional focus	4.50	3.25	3.67	3.83
Narrative presence	3.75	3.67	3.00	3.25
Emotional engagement	3.83	3.25	3.17	3.50
Irritable	3.25	2.75	2.50	2.33
Distressed	2.83	2.17	2.08	2.50
Afraid	2.75	1.92	2.33	2.08
Nervous	2.92	1.92	2.83	2.50
Scared	2.67	2.17	1.67	2.25
Upset	2.08	1.92	2.33	2.42
Excited	2.50	2.50	2.67	2.25
Interested	3.58	2.17	2.67	3.33
Alert	2.17	2.67	2.42	2.50

Table 1: Mean scores of study results (red cells are 3.0 or higher)

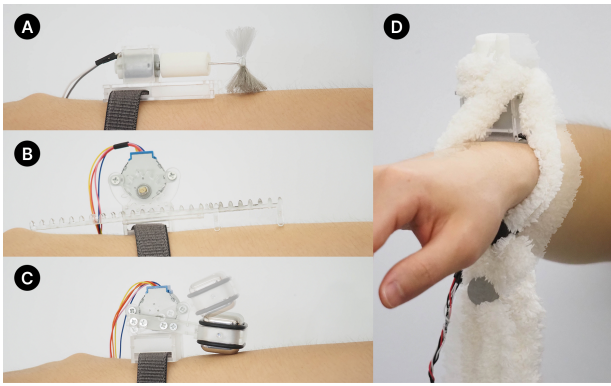


Figure 2: Movements of haptic devices

experience for DHH audiences [9, 15]. In particular, our main focus is on investigating tactile modalities that move beyond widely studied vibration-based approaches [3, 10, 11], aiming to examine the untapped potential of these alternative channels for horror media accessibility.

This project proposes four novel wrist-worn tactile devices designed to help DHH audiences better immerse themselves in horror films through embodied sensory experiences [7, 19]. Rather than attempting to replicate or replace sound, our system translates the tension-and-release dynamics of horror movies into four distinct tactile sensations: *texture*, *locomotion*, *temperature*, and *pressure*. To understand how these tactile approaches can make horror films more accessible, we conducted a user study with 12 DHH participants, evaluating each prototype’s ability to enhance narrative engagement and trigger emotional responses. Our work contributes to creating more accessible media experiences by exploring how diverse tactile modalities can enhance cinematic experiences for DHH audiences, offering insights for more inclusive technologies.

2 Silent suspense

Silent Suspense consists of four different devices, each designed to replicate common fear sensations from everyday life [6]. The *Texture* device simulates aggressive scratching, *Locomotion* mimics

something crawling on the skin, *Temperature* recreates the chill of goosebumps, and *Pressure* simulates being seized tightly.

Texture device features a circular mohair cylinder on a DC motor (controlled by L298N driver) positioned to lightly touch the arm. It spins slowly during suspense-building moments for subtle roughness, and rapidly during climactic scenes for intense scratching sensations (Figure 2 A). **Locomotion** device uses a 28BY-48 stepper motor to move an acrylic rod back and forth across the arm. At the end of the rod, there is a projection that traces over the skin during movement. The movement starts slow and gentle during suspense-building moments, then accelerates during climactic scenes to create sudden tickling sensations (Figure 2 B). **Temperature** device features a stainless steel ice cube housed in a holder. The holder is attached to a stepper motor that moves the ice up and down. It lightly taps the arm during suspense-building moments causing a subtle chill, then presses firmly during climactic scenes for stronger cold sensations (Figure 2 C). **Pressure** device uses wool yarn threaded through a cylindrical component attached to a stepper motor. The yarn tightens around the wrist, starting gently during suspense-building moments and becoming intense during climactic scenes (Figure 2 D).

3 User study

12 DHH participants were recruited through a community welfare center. Each received 40 USD for their 120-minute participation. Participants were organized into four groups of three and each group was supported by a sign language interpreter. The study consisted of four sessions. In each session, all four groups watched a 6-minute horror film with each group wearing a different device. They then completed questionnaires rating their experience. Groups rotated clockwise between sessions to ensure all participants experienced every device. The study concluded with follow-up interviews.

3.1 Results

To understand the effectiveness of the four Silent Suspense devices in enhancing accessible horror film experiences, potential applications, and implementation approaches, both quantitative and qualitative data were collected. Quantitative assessment used questionnaires with 13 questions rated on 5-point Likert scales, selected from the Narrative Engagement Scale [4] and the Positive

and Negative Affect Schedule (PANAS) Scale [17]. Using 3 as a neutral standard, we identified items with means of 3 or higher as indicators of meaningful user experience for each device. For qualitative evaluation, we analyzed audio recordings from each group's 20-minute interview.

Quantitative Results. All four devices demonstrated effectiveness across narrative engagement measures, with mean scores of 3 or higher (top 4 items in 1). Meanwhile, their influence on PANAS measures was comparatively limited, with only three items exceeding the neutral standard (bottom 9 items in 1). For narrative engagement, both *Texture* and *Pressure* helped people understand the film's narrative better (4.17 ± 0.69 and 4.17 ± 0.80 , respectively). *Texture* also led to higher attentional focus (4.50 ± 0.65), with *Pressure* and *Temperature* also proving effective (3.83 ± 1.07 and 3.67 ± 1.18 , respectively). Additionally, *Locomotion* enhanced narrative presence (3.67 ± 1.03). Regarding PANAS, only *Texture* triggered both irritability (3.25 ± 1.16) and interest (3.58 ± 1.44), while *Pressure* also generated interest (3.33 ± 1.31). Overall, the findings indicate that *Texture* proved to be the most effective tactile modality for enhancing narrative engagement and emotional immersion in horror films for DHH audiences.

Qualitative Findings. Analysis of the interview recordings revealed two main themes: application potential and design considerations for tactile devices. Participants suggested expanding beyond horror films to other genres, such as tickling sensations for romance films or strong tightening for explosions in action films. Also, they proposed practical applications like danger recognition and daily emergency alerts. Participants further emphasized the importance of precise timing and intensity control. Other recommendations included utilizing gradual operation to build suspense, mentioning the *Pressure*'s effective gradual increase in intensity. This feedback highlights the need for sophisticated intensity mapping that adapts to different narrative phases, thereby enhancing the viewing experience without overwhelming it.

4 Conclusion

This study explored diverse tactile modalities to enhance horror film experiences for DHH audiences using four novel tactile-based prototypes representing *texture*, *locomotion*, *temperature*, and *pressure* sensations. Findings showed all tactile modalities enhanced narrative engagement, with *Texture* performing most effectively overall. This work contributes to developing accessible media experiences through tactile interactions, opening new possibilities for cinematic engagement that goes beyond audiovisual elements to provide DHH audiences with richer horror film experiences.

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